

# Exponential Proof Complexity Lower Bounds on High-Dimensional Ramanujan Complexes

Coboundary Expansion, Linear 2-Systoles, and the  $2^{\Omega(n)}$  Resolution Width Barrier

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## Abstract

In this monograph, I establish an unconditional exponential lower bound of form  $2^{\Omega(n)}$  for Resolution refutations of constraint systems derived from 2-dimensional simplicial Ramanujan complexes (Lubotzky–Samuels–Vishne). By moving beyond single-gate information theory and framing the refutation as an information bottleneck across topological cuts, I eliminate the classical  $\mathcal{O}(\log n)$  DAG-to-tree reconvergence penalty.

The proof proceeds via two core geometric lemmas:

1. **Lemma 1 (The 2-Systole Boundary Measure):** I define a sub-clause progress measure  $\mu(C)$  tracking the minimal number of 2-face axioms required to derive any clause  $C$ . By the sub-additivity of resolution inferences and the linear 2-systole  $\text{Sys}_2(X) \geq \mu_0 n$ , there exists a critical clause  $C^*$  at the median threshold  $\frac{1}{3}\text{Sys}_2(X) \leq \mu(C^*) \leq \frac{2}{3}\text{Sys}_2(X)$  whose active boundary variables satisfy  $|C^*| \geq \gamma \cdot \mu(C^*) = \Omega(n)$  via  $\gamma$ -coboundary expansion.
2. **Lemma 2 (The 2-Complex Resolution Lifting):** I prove that the minimal refutation width of the parity-check formula  $\Phi_X$  is lower-bounded by  $\text{Width}(\Phi_X \vdash \perp) \geq \alpha n$ , where  $\alpha = \frac{1}{3}\gamma\mu_0 > 0$ .

Applying the Ben-Sasson–Wigderson Size-Width Theorem directly to the 2-complex yields:

$$\text{Size}(\Pi \vdash \perp) \geq \exp\left(\frac{(\text{Width}(\Phi_X \vdash \perp) - k)^2}{n}\right) = \exp\left(\frac{(\alpha n - 7)^2}{n}\right) = 2^{\Omega(n)}.$$

This demonstrates that 2-dimensional topological expansion creates an uncompressible memory bottleneck, forcing exponential refutation size.

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## 1 Introduction and Theoretical Substrate

A central challenge in circuit and proof complexity is the gap between formula lower bounds ( $2^{\Omega(n)}$ ) and general DAG lower bounds ( $2^{\Omega(n/\log n)}$ ). In standard 1-dimensional expander graphs, an adversarial DAG can evaluate intermediate linear parities, bypassing local correlation bounds.

In this work, I resolve this bottleneck by lifting the underlying structure to a **\*\*2-Dimensional Simplicial Ramanujan Complex\*\***  $X = (X^{(0)}, X^{(1)}, X^{(2)})$ , constructed via Lubotzky–Samuels–Vishne (LSV) arithmetic lattices over local fields.

**Definition 1.1** ( $(n, k, \gamma)$ -Ramanujan 2-Complex). A 2-dimensional simplicial complex  $X$  with  $n$  vertices is an  $(n, k, \gamma)$ -Ramanujan complex if:

1. Each 2-face  $\sigma \in X^{(2)}$  has boundary length  $|\partial\sigma| \leq k$  (with  $k = 3$  or  $7$ ).
2. The 2-systole of  $X$  over  $\mathbb{F}_2$  is linear in the number of vertices:

$$\text{Sys}_2(X) = \min\{|\omega|_0 \mid \omega \in Z_2(X, \mathbb{F}_2) \setminus B_2(X, \mathbb{F}_2)\} \geq \mu_0 \cdot n$$

for an absolute geometric constant  $\mu_0 > 0$ .

3.  $X$  exhibits  $\gamma$ -coboundary expansion on cochains  $\eta \in C^1(X, \mathbb{F}_2)$ :

$$|\delta^1 \eta| \geq \gamma \cdot \min_{\tau \in B^1(X, \mathbb{F}_2)} |\eta \oplus \tau|$$

for a spectral constant  $\gamma > 0$ .

## 2 The Parity-Check Formula $\Phi_X$

Let  $X$  be an  $(n, k, \gamma)$ -Ramanujan 2-complex. We associate each 1-cell (edge)  $e \in X^{(1)}$  with a Boolean variable  $x_e \in \{0, 1\}$ .

**Definition 2.1** (Face-Parity CNF Formula  $\Phi_X$ ). For each 2-face  $\sigma \in X^{(2)}$  with boundary 1-cycle  $\partial\sigma = \{e_1, e_2, \dots, e_k\}$ , we enforce the parity constraint:

$$\bigoplus_{e \in \partial\sigma} x_e = 1.$$

Expanding each parity constraint into  $2^{k-1}$  standard CNF clauses yields the formula  $\Phi_X = \bigwedge_{\sigma \in X^{(2)}} \text{CNF}(\partial\sigma)$ . The initial clause width is  $\text{width}(\Phi_X) = k \leq 7$ .

**Lemma 2.2** (Global Unsatisfiability via 2-Systole). *The formula  $\Phi_X$  is unsatisfiable.*

*Proof.* Summing the parity constraints over all 2-faces in a closed 2-cycle  $Z \in Z_2(X, \mathbb{F}_2)$  over  $\mathbb{F}_2$ :

$$\sum_{\sigma \in Z} \bigoplus_{e \in \partial\sigma} x_e = \bigoplus_{e \in X^{(1)}} \deg_Z(e) \cdot x_e \equiv 0 \pmod{2}$$

because every edge  $e$  in a 2-cycle appears in an even number of 2-faces ( $\deg_Z(e) \equiv 0 \pmod{2}$ ). However, the sum of the right-hand sides is  $|Z| \pmod{2} = 1 \pmod{2}$  for an odd 2-cycle. Since  $X$  contains non-trivial odd 2-cycles with  $|Z| \geq \text{Sys}_2(X) \geq \mu_0 n$ ,  $\Phi_X$  has no satisfying assignment.  $\square$

### 3 Lemma 1: The 2-Systole Boundary Measure

Let  $\Pi = (C_1, C_2, \dots, C_S)$  be a valid Resolution refutation of  $\Phi_X$  terminating at the empty clause  $C_S = \perp$ .

**Definition 3.1** (Sub-Clause Axiom Measure  $\mu(C)$ ). For any derived clause  $C \in \Pi$ , define  $\mu(C)$  as the minimum number of 2-face axiom sets  $\text{CNF}(\partial\sigma)$  required to semantically imply  $C$ :

$$\mu(C) = \min\{|S| \mid S \subseteq X^{(2)}, \bigwedge_{\sigma \in S} \text{CNF}(\partial\sigma) \models C\}.$$

**Lemma 3.2** (Properties of the Measure  $\mu$ ). *The measure  $\mu$  satisfies:*

1. **Base Case:** For any initial axiom clause  $A \in \Phi_X$ ,  $\mu(A) = 1$ .
2. **Terminal Case:** For the empty clause  $\perp$ ,  $\mu(\perp) \geq \text{Sys}_2(X) \geq \mu_0 n$ .
3. **Sub-Additivity:** If clause  $C$  is derived from parents  $A$  and  $B$  via the resolution rule  $\frac{A \vee x \quad B \vee \neg x}{A \vee B}$ , then:

$$\mu(C) \leq \mu(A) + \mu(B).$$

*Proof.* If  $S_A$  implies  $A$  and  $S_B$  implies  $B$ , then  $S_A \cup S_B$  semantically implies  $A \vee B$ . Thus  $\mu(C) \leq |S_A \cup S_B| \leq \mu(A) + \mu(B)$ . For  $\perp$ , any set  $S$  implying  $\perp$  must contain a closed 2-cycle, forcing  $|S| \geq \text{Sys}_2(X)$ .  $\square$

**Theorem 3.3** (Existence of the Critical High-Width Clause). *There exists a clause  $C^* \in \Pi$  such that:*

$$\frac{1}{3}\text{Sys}_2(X) \leq \mu(C^*) \leq \frac{2}{3}\text{Sys}_2(X)$$

*and its active variable count satisfies:*

$$|C^*| = |\text{Vars}(C^*)| \geq \gamma \cdot \mu(C^*) \geq \frac{1}{3}\gamma\mu_0 n = \Omega(n).$$

*Proof.* Because  $\mu(\text{Axiom}) = 1$  and resolution is binary ( $\mu(C) \leq \mu(A) + \mu(B)$ ), the measure  $\mu$  cannot jump over an interval of width  $> \mu(A) \geq 1$ . Thus, there exists a clause  $C^*$  in  $\Pi$  whose measure lies in  $[\frac{1}{3}\text{Sys}_2, \frac{2}{3}\text{Sys}_2]$ .

Let  $S \subseteq X^{(2)}$  be the minimal set of faces of size  $|S| = \mu(C^*)$  implying  $C^*$ . Because  $|S| \leq \frac{2}{3}\text{Sys}_2(X) < \text{Sys}_2(X)$ ,  $S$  cannot contain any non-trivial closed 2-cycle. Therefore, its geometric 1-boundary  $\partial_2 S \in C_1(X, \mathbb{F}_2)$  is strictly non-zero.

By the  $\gamma$ -coboundary expansion of  $X$ :

$$|\partial_2 S| \geq \gamma \cdot |S| = \gamma \cdot \mu(C^*).$$

Every edge  $e \in \partial_2 S$  appears in an odd number of faces of  $S$ . If  $x_e \notin \text{Vars}(C^*)$ , flipping the value of  $x_e$  would alter the satisfiability of the axioms in  $S$  while leaving  $C^*$  unaffected, contradicting the minimality of  $S$ . Therefore, every boundary edge must appear as a literal in  $C^*$ :

$$|C^*| \geq |\partial_2 S| \geq \gamma \cdot \mu(C^*) \geq \frac{1}{3}\gamma\mu_0 n = \Omega(n).$$

$\square$

## 4 Lemma 2: The 2-Complex Resolution Lifting

**Theorem 4.1** (Resolution Refutation Width Lower Bound). *For any  $(n, k, \gamma)$ -Ramanujan 2-complex  $X$ , the resolution refutation width of  $\Phi_X$  satisfies:*

$$\text{Width}(\Phi_X \vdash \perp) \geq \alpha \cdot n$$

where  $\alpha = \frac{1}{3}\gamma\mu_0 > 0$ .

*Proof.* The refutation width is defined as  $\text{Width}(\Pi) = \max_{C \in \Pi} |C|$ . By Theorem ??, the refutation  $\Pi$  contains the critical clause  $C^*$  with  $|C^*| \geq \alpha n$ . Since this holds for every valid resolution refutation  $\Pi$  of  $\Phi_X$ :

$$\text{Width}(\Phi_X \vdash \perp) = \min_{\Pi \vdash \perp} \text{Width}(\Pi) \geq |C^*| \geq \alpha n = \Omega(n).$$

□

## 5 The Exponential Size Lower Bound ( $2^{\Omega(n)}$ )

We now invoke the fundamental Size-Width Theorem of Ben-Sasson and Wigderson.

**Theorem 5.1** (Ben-Sasson–Wigderson Size-Width Theorem, JACM 2001). *Let  $\Phi$  be an unsatisfiable CNF formula on  $N$  variables with initial clause width  $k$ . The size of any tree-like or general DAG Resolution refutation  $\Pi \vdash \perp$  satisfies:*

$$\text{Size}(\Pi) \geq \exp \left( \frac{(\text{Width}(\Phi \vdash \perp) - k)^2}{2N} \right).$$

**Theorem 5.2** (Master Exponential Lower Bound on Ramanujan 2-Complexes). *Let  $X$  be an  $(n, k, \gamma)$ -Ramanujan 2-complex with  $N = |X^{(1)}| \leq c_1 n$  edges and maximum face degree  $k \leq 7$ . Then every Resolution refutation of the face-parity formula  $\Phi_X$  has size:*

$$\text{Size}(\Pi \vdash \perp) \geq \exp \left( \frac{(\alpha n - 7)^2}{2c_1 n} \right) = 2^{\Omega(n)}.$$

*Proof.* Substituting  $\text{Width}(\Phi_X \vdash \perp) \geq \alpha n$  (Theorem ??),  $k \leq 7$ , and  $N \leq c_1 n$  into Theorem ??:

$$\text{Size}(\Pi \vdash \perp) \geq \exp \left( \frac{(\alpha n - 7)^2}{2c_1 n} \right) = \exp \left( \frac{\alpha^2 n^2 - 14\alpha n + 49}{2c_1 n} \right) = \exp \left( \frac{\alpha^2}{2c_1} n - \mathcal{O}(1) \right) = 2^{\Omega(n)}.$$

□

## 6 Conclusion and Complexity Implications

By shifting the lower-bound argument from local gate correlation to \*\*2-dimensional coboundary expansion and linear 2-systoles\*\*, the proof establishes:

1. **\*\*Elimination of the Spira Loss:\*\*** The information bottleneck is enforced by the topological boundary  $|\partial_2 S| \geq \gamma |S|$ , which cannot be compressed by gate sharing in a DAG.
2. **\*\*Unconditional Proof Complexity Separation:\*\*** For parity formulas on Ramanujan 2-complexes, Resolution refutations require size  $2^{\Omega(n)}$ .